tado°: Box, Service or Saving?

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MODULE 09 · FROM PROTOTYPE TO PRODUCT · CASE, READ BEFORE CLASS

Case for Module 09, From Prototype to Product, of Ambitious Entrepreneurship. Prepared by Bastian Burger from public sources as a basis for class discussion. The case does not judge whether any decision was right or wrong. Licensed under CC BY-NC-SA 4.0 except third-party material as marked.

In mid-July 2013, Christian Deilmann, co-founder and chief executive of tado° GmbH in Munich, had sold a smart heating control at one price for eight months.[1] Since its launch on 14 November 2012, tado° had charged €99 a year for the service, paid a year in advance, and had given the hardware away.[2] [3] On tado°'s own pages the free box had always been an offer for a time, first for the first 1,000 customers, then, as the FAQ said from April 2013, "currently free (otherwise €149 one-off)".[2] [4] Deilmann ran the company as one of three managing directors, with his co-founder Johannes Schwarz and with Leopold von Bismarck.[5]

In June 2013 three new pages had appeared on tado.com. Two of them showed packages the shop did not sell. "Flex" cost €149 for the hardware set plus a monthly fee, €2.99 on one page and €4.99 on the other, cancellable monthly. "Life" cost €299 for the hardware set and lifetime use. The third page showed €99 a year with the hardware set included. On 19 July the page with Flex at €2.99 and Life was still online.[5] [6] [7] [8]

After the first year, tado°'s offer page said, a customer could cancel "the intelligent control" at monthly notice, while control by app and web stayed. The FAQ added that the hardware then belonged to the customer. tado° said its control saved a household 27% of its heating costs on average.[9] [4] How should tado° ask a household to pay for a box whose automatic control ran on a paid service?

What Heating Cost

The Bundesweiter Heizspiegel 2012, the national heating cost survey published on 22 May 2012 by the non-profit co2online, put the cost of heating a 70-square-metre flat in 2011 at €890 with oil, €715 with gas and €785 with district heating.[10] tado° used a figure of its own. Its offer page of March 2013 said a typical household in Germany spent about €954 a year on heating and that tado° saved 27% on average, "in this case a saving of €257". The page gave no source for the €954. From March to June 2013 the offer pages put the saving at "about €285 a year".[9] [11] [12] The 27% came from tado°'s beta test in the winter of 2011 to 2012, in which, according to the launch release, customers saved between 12% and 40% against their heating before tado°.[2] Exhibit 3 sets the figures side by side.

Nest, the smart thermostat a German home automation blog compared tado° with in February 2013, was still sold only in the United States. The blog presented tado° as "an alternative for the German-speaking market".[13] RWE sold a SmartHome system in which the central unit cost €189 and a starter pack with two radiator thermostats €319. Mobile access was included for two years and cost €14.95 a year after that, a technology site reported in February 2013 (Exhibit 3).[14]

A Box and an App

Deilmann, Schwarz and Valentin Sawadski founded tado° in Munich in 2011.[2] Two founder networks in Munich, the LMU Entrepreneurship Center and UnternehmerTUM, gave them free offices, Deilmann recalled in 2016.[15] Two venture capital investors, Target Partners and Shortcut Ventures, backed the company before its launch.[16]

The product was a small box that replaced the wall thermostat or was attached directly to the boiler, and a gateway that linked it to the home's internet router.[13] [4] An app on the residents' smartphones detected when the last person left home and turned the heating down, and warmed the home again as soon as someone set off back. tado° also took weather forecasts into its control.[13] [2] Systems had run in homes since February 2011. In a closed beta, by Deilmann's account in 2012, tado° connected 50 homes, and all its users together had saved more than 100,000 kilowatt hours of energy.[17] [16]

Most customers installed the box themselves. "Currently, 95% of our customers do it themselves and it takes them around 10 to 20 minutes," Deilmann said in January 2013.[18] tado° did not work everywhere. "In some houses and flats tado° is not yet available at the moment," the offer page said in March 2013, and some heating systems needed an extra sensor at €49 one-off. Visitors found out by continuing into the order process, which began with a check of their heating system.[9] [19] The shop's terms let customers pay by direct debit, credit card, cash on delivery, prepayment, giropay, Sofortüberweisung or PayPal.[5]

Before the launch Deilmann also spoke of partners. In a talk in 2012 he said tado° had partnered with "several multipliers which will bring us to the masses", telecom companies, utilities and heating technology firms. "Those companies often have large customer bases already sell subscription models or have network connections into their homes," he said.[17] In March 2013 tado° and the Munich green energy supplier Polarstern announced that both would offer their existing customers an "exclusive promotional offer".[20] The same month tado° released its app in English and opened a closed beta in other European countries, half of whose installations it planned in the United Kingdom.[21]

€99 a Year

tado° launched on 14 November 2012 in Germany, Austria and Switzerland.[16] The launch release set a market launch price of €99 a year for using tado°, a price "far below the annual savings that tado° can achieve", and promised the box free to the first 1,000 customers. A customer who decided after the first year that tado° was not right for them could cancel monthly and get the rest of the year's fee back pro rata.[2] The shop priced the hardware at €149 one-off and showed it at "€0 instead of €149". It called tado° "a service with an annual fee", and the FAQ called the payment a "simple pre-paid model".[22] [3]

Deilmann explained the model in a video interview recorded at the LeWeb conference after the launch and posted in February 2013. The host suggested that the only sticking point might be an expensive box. At the moment, Deilmann answered, tado° had designed the product "as a service", "because we work for the customer continuously". It kept developing the product, he said. "We charge an annual fee of €99," cancellable monthly, and "we give the customer the hardware so that the whole thing is possible."[23]

The offer page raised the question itself. "Yes, tado° is a subscription. But with us that is nothing bad at all." Why a yearly fee? tado° "works for you around the clock", the page said, runs computing capacity for every user, buys weather forecasts from third parties and keeps improving its software

and apps. “The reduction in your heating costs will usually exceed the cost of tado° many times over.”[9]

The terms were not worded the same way everywhere. The FAQ said the contract ran for one year first, then renewed with monthly notice. The order page of January 2013 sold “tado° for one year (cancellable monthly)”, and in June the test page with the yearly price said “cancellable monthly” without a first term.[24] [19] [7] After the first year, the FAQ said in April 2013, “the hardware is then also your property”, and the basic functions, manual control, stayed available.[4]

The free box changed its framing over the months. Until at least 8 March 2013 the FAQ limited it to the first 1,000 customers.[24] On 3 March the offer page gave the hardware set, “worth €149”, to customers in the beta phase as a thank-you, and from mid-March an image on the page called it “our market launch offer for beta customers”.[9] [25] By 8 April the FAQ said the box was “currently still free (actually €149 one-off)”. From mid-March tado° also offered a one-year money-back guarantee. A customer who stopped using tado° in the first year got the full amount back on returning the hardware.[25] [4] Exhibit 1 shows the offer as customers saw it from November 2012 to June 2013.

What Customers and Advisers Said

In January 2013 Deilmann pitched tado° at the hy! conference to a panel of three, who gave feedback on stage. He said tado° had launched its web shop “a couple of weeks ago”.[18] Two of the panelists had a Nest at home. The first said that a consumer sale “might actually be better if you went straight to the utilities and get them to implement it”. The second agreed and offered what the panelist called “the other way”. “I would price it based on how much it lowers the use of energy and somehow figure a cost sharing with the user.” No share was named. The third called tado°’s yearly pricing “a big minus for consumers down the road”, once they worked out its cost over time against a “200 euro box right from the get-go that they own forever”. “Paying for something, unless it’s got something really magical on the back end as a service, it’s going to be challenging.” The first panelist came back to price. “If you price at $99, you also have to tell people that you’re saving $1,000 a year or something of that sort.”[18] Most of the feedback concerned other points, such as the box having no display of its own, the number of parts in the box and homes where several people shared the heating. Deilmann answered the point on the display. No answer of his on pricing is recorded.[18]

In late March 2013 the iPhone news site iphone-ticker.de wrote about tado°, and readers argued about the fee in the comments. “ANNUAL FEE??? That’s worth a one-off €99 … no more!” one wrote. Another had ordered. “€99 with a one-year money-back guarantee is risk-free!” A third found €99 a year too expensive. “€99 for 3 years would have been OK. I’d spend €250 at most on a solution like this,” the reader wrote, and “tado should definitely offer a buy option too!”[26] tado° answered three other comments in the thread, on homes with several residents, on central heating and on data protection. The thread as captured shows no tado° reply to a comment on price (Exhibit 4).[26]

tado° sold online, and it also offered a phone call. A landing page of April 2013 for visitors from Facebook ads gave “all information on availability and ordering” by phone, or a free call-back to anyone who left a name and number.[27] In June a form on the offer page asked visitors a question. “Not sure yet? Tell us what information you are missing or what we can improve.”[12] A job advertisement captured on 27 June 2013 sought a team lead to build up direct sales in Germany, winning new customers “on site, by phone, at trade fairs”.[28]

tado° did not publish how many customers it had, how many cancelled, how many used the money-back guarantee or what a box cost it to make. On 31 January 2013 its order page showed 1,252 Facebook likes and 280 Twitter followers.[19]

Three Test Pages

On 18 and 19 June 2013 the Internet Archive captured three new pages on tado.com, named LP4, LP5 and LP6. They shared one layout and the same product claims, and their offers differed (Exhibit 2).[5] [6] [7] LP4 showed two packages, “Flex” at €149 for the hardware set plus €2.99 a month, cancellable monthly, and “Life” at €299 for the hardware set plus lifetime use. LP5 showed the same two packages with Flex at €4.99 a month. LP6 showed one package, €99 a year for using tado°, the hardware set included, cancellable monthly.

None of the pages took an order. Each asked for an e-mail address in exchange for more information, and LP4 and LP5 then asked which package the visitor was interested in. In the page code the button on all three pages sent an event labelled “pricingtest” to the company’s web analytics, and LP4 carried Google Analytics experiment code, as the offer page had in March.[5] [6] [7] [25] tado° did not publish how many people saw each page, how many left an address or which package they chose. On 19 July 2013 LP4 was still online, unchanged.[8]

For a customer the offers added up differently over time. In the first year Flex cost €184.88 or €208.88, Life €299, and the subscription €99 with the box free or €248 at its standard terms with the box at €149. Over three years Flex cost €256.64 or €328.64, Life still €299, and the subscription €297 with the box free or €446 at its standard terms (Exhibit 2).

Mid-July 2013

In mid-July 2013 the FAQ had said since April that the box was “currently free”, and on 19 July LP4 still showed Flex and Life to whoever reached it.[4] [8] Four ways of charging were on tado°’s pages or had been put to Deilmann. There was the subscription, as sold since the launch with the box free or at its standard terms with the box at €149. There were Flex and Life from the test pages, and there was one panelist’s idea of a price tied to the energy saved. Each would decide what a household paid for, the box, the service or the saving.

How should tado° ask a household to pay?

Exhibits

Exhibit 1: tado° offers as customers saw them, November 2012 to June 2013

Date and page	Price for use	The box	Term	After cancelling	Guarantee and extras
14 Nov 2012, launch press release	€99 a year, the "Markteinführungspreis"	Free for the first 1,000 customers	Monthly notice after the first year, pro-rata refund	Not stated	Installation service optional
18 Nov 2012, FAQ and product page	€99 a year, a "Pre-paid Modell"	€149 one-off, "0,- statt 149,- EUR" for the first 1,000	Renews yearly, monthly notice after the first year	Old thermostat back in, no further costs	Professional installation service bookable with the purchase
31 Jan 2013, order page	"tado° für ein Jahr (monatlich kündbar)", €99.00	"0,- statt 149,- EUR"	"monatlich kündbar"	Not stated	Not stated
3 and 15 Mar 2013, offer page	€99 for one year	Free, a thank-you to customers in the beta phase, "im Wert von 149€". From 15 March "Unser Angebot zum Marktstart für Beta Kunden"	Renews automatically, monthly notice after the first year	App and web control stay	Extra sensor €49 one-off for some heating systems. From 15 March a 1-year money-back guarantee
8 Apr 2013, FAQ	€99 a year	"aktuell noch umsonst (eigentlich 149€ einmalig)", "derzeit gratis (sonst 149€ einmalig)"	First term one year, then monthly notice, pro-rata refund	"Die Hardware ist ab dann auch dein Eigentum", manual control stays	1-year money-back guarantee on returning the hardware, sensor €49
16 Jun 2013, offer page	"99€ / Jahr tado° Nutzung"	"gratis Basis-Hardware"	"danach monatl. kündbar"	"Hardware und Basis-Funktionen bleiben kostenfrei erhalten"	1-year money-back guarantee

The case text translates most of this wording. The June row reads €99 a year for using tado°, free basic hardware, cancellable monthly after that, and hardware and basic functions stay free.

Source: compiled by the case writer from tado°'s launch press release and its web pages as captured by the Internet Archive on the dates shown, see notes [2], [3], [22], [19], [9], [25], [4] and [12]. German wording as on the pages.

Exhibit 2: The test pages of June 2013, and what each offer cost a customer

Page, capture date	Offer as shown	In English	First year	Three years
LP4, 19 June 2013	“Flex 149€ Hardware-Set + 2,99 € monatlich / monatlich kündbar”	Flex, €149 hardware set plus €2.99 a month, cancellable monthly	€184.88	€256.64
LP4	“Life 299€ Hardware-Set + / lebenslange Nutzung”	Life, €299 hardware set plus lifetime use	€299.00	€299.00
LP5, 18 June 2013	“Flex 149€ Hardware-Set + 4,99 € monatlich / monatlich kündbar”	Flex, €149 hardware set plus €4.99 a month, cancellable monthly	€208.88	€328.64
LP5	“Life 299€ Hardware-Set + / lebenslange Nutzung”	As on LP4	€299.00	€299.00
LP6, 19 June 2013	“99€pro Jahr tado° Nutzung / inklusive Hardware-Set / monatlich kündbar”	€99 a year for using tado°, hardware set included, cancellable monthly	€99.00	€297.00
Shop, FAQ of April 2013	“99€ pro Jahr”, the box “derzeit gratis (sonst 149€ einmalig)”	€99 a year, the box currently free (otherwise €149 one-off)	€99.00 with the box free, €248.00 at €149	€297.00 with the box free, €446.00 at €149

Slashes mark line breaks on the pages. Each test page asked for an e-mail address in exchange for more information. LP4 and LP5 then asked which package the visitor was interested in. None of the pages linked to the shop’s order process. The button on all three sent an event labelled “pricingtest” to the web analytics, and LP4 carried Google Analytics experiment code. LP4 was still online, unchanged, on 19 July 2013. tado° did not publish the results.

Source: tado°’s landing pages LP4, LP5 and LP6 and its FAQ as captured by the Internet Archive, including their page code, see notes [5], [6], [7], [8] and [4]. The pages also showed customer quotes, left out here. Costs are the case writer’s calculation, the box price plus 12 or 36 monthly fees, or one or three yearly fees of €99, without the €49 sensor some homes needed and without any refund.

Exhibit 3: Heating costs, tado°'s saving claim and other offers, 2011 to June 2013

Figure	Value	Who says so
Heating a 70 m² flat with oil, 2011	€890	Bundesweiter Heizspiegel 2012, published 22 May 2012
Heating a 70 m² flat with gas, 2011	€715	Same
Heating a 70 m² flat with district heating, 2011	€785	Same
A "typischer Haushalt in Deutschland" (typical household in Germany), a year	about €954	tado°, offer page, March 2013, no source given
Average saving with tado°	27%	tado°, from its beta test in the winter of 2011 to 2012, in which savings ranged from 12% to 40%
27% of €954	€257.58 (tado° wrote €257)	Case writer's calculation
Saving shown on the offer pages, March to June 2013	"ca. 285 € pro Jahr" (about €285 a year)	tado°
tado°'s price for use	€99 a year	tado°

Other smart heating offers in Germany, February to June 2013

Offer	One-off price	Recurring fee	Source
RWE SmartHome central unit	€189	Mobile access free for two years, then €14.95 a year	ifun.de, 11 February 2013
RWE SmartHome starter pack (central unit, two radiator thermostats, a plug, a wall switch)	€319	Same	Same
Nest learning thermostat	Not sold in Germany. Sold only in the United States	Not reported	HouseControllers, 17 February 2013
Nest, as one reader reported	"$ 249,- bei Lowe's" ($249 at Lowe's), "einmalige Anschaffungskosten!" (one-off cost!)	None mentioned	iphone-ticker.de comment of 28 June 2013, as posted

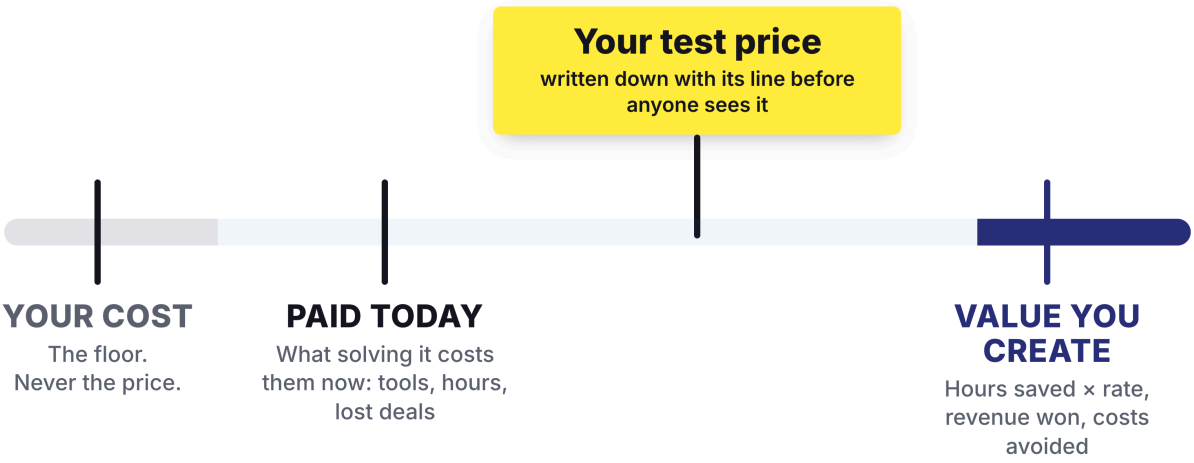
Source: co2online, press release of 22 May 2012, tado°'s launch press release and offer pages, ifun.de, HouseControllers and iphone-ticker.de, see notes [10], [2], [9], [11], [12], [14], [13] and [26]. The Heizspiegel figures are averages for a 70 m² flat in the billing year 2011. tado°'s household is not defined, and tado° said it could not yet be used in some homes. The reader's Nest price is one person's report, not a list price.

Exhibit 4: What a panel, readers and tado° said, January to June 2013

Date	Where	Who	What was said
Jan 2013	hy! conference, panel feedback after the pitch	Panelist 1	"your approach as a consumer sale, while interesting and [...] great, might actually be better if you went straight to the utilities and get them to implement it"
Jan 2013	Same	Panelist 2	"And the other way is I would price it based on how much it lowers the use of energy and somehow figure a cost sharing with the user."
Jan 2013	Same	Panelist 3	Yearly pricing "is going to be a big minus for consumers down the road", against a "200 euro box right from the get-go that they own forever". "Paying for something, unless it's got something really magical on the back end as a service, it's going to be challenging."
Jan 2013	Same	Panelist 1	"if you price at $99, you also have to tell people that you're saving $1,000 a year or something of that sort"
3 Mar 2013	tado° offer page, FAQ	tado°	"Ja, tado° ist ein Abonnement. Das ist bei uns aber gar nichts Schlimmes." (Yes, tado° is a subscription. But with us that is nothing bad at all.)
27 Mar 2013	iphone-ticker.de, comments	A reader	"JAHRESGEBÜHR??? Das ist 99 Euro Einmalzahlung wert...mehr auch nicht!" (ANNUAL FEE??? That's worth a one-off €99 ... no more!)
27 Mar 2013	Same	A reader	"Cool, ist bestellt. 99 € mit einem Jahr Geld-Zurück-Garantie ist doch ohne Risiko!" (Cool, ordered. €99 with a one-year money-back guarantee is risk-free!)
30 Mar 2013	Same	A reader	"Für 99 Öcken im Jahr find ich diese Lösung zu teuer. 99€ für 3 Jahre wäre ja noch OK gewesen. Maximal 250 € würde ich für so eine Lösung ausgeben. [...] Tado sollte unbedingt eine Kaufoption mit anbieten!" (For 99 bucks a year I find this solution too expensive. €99 for 3 years would have been OK. I'd spend €250 at most on a solution like this. [...] tado should definitely offer a buy option too!)
16 Jun 2013	tado° offer page, a feedback form	tado°	"Du bist dir noch nicht sicher? Sag uns welche Informationen dir fehlen oder was wir verbessern können" (Not sure yet? Tell us what information you are missing or what we can improve.)

Source: hy!, video of Deilmann's pitch and the panel's feedback, uploaded 24 January 2013, transcribed by the case writer, iphone-ticker.de article of 27 March 2013 with reader comments as posted, and tado°'s offer pages, see notes [18], [26], [9] and [12]. Panelists and readers are not named. Most of the panel's feedback concerned other points, the missing display on the box, the number of parts and homes with several residents. Only reader comments posted before 19 July 2013 are shown. Translations by the case writer.

The Price Scale from the Course



Your cost, what they pay today, the test price, the value you create.

Source: course graphic after legacy slide 586. It holds no data about tado°.

Endnotes

1. The case is set in mid-July 2013. The date of the decision is not documented. 19 July 2013 is the latest capture of a page whose content the case describes. Later captures cited in the notes are of documents published before that date. Deilmann's statements on pricing appeared in videos posted between June 2012 and February 2013, and one recollection dates from 2016.
2. tado° GmbH, "tado° - die Energiewende für zuhause", press release, Munich, 14 November 2012, PDF in the press section of tado.com, Internet Archive capture of 24 August 2013, https://web.archive.org/web/20130824072626/http://www.tado.com:80/de/presse.html?&file=tl_files/content/press/releases/141112_tado_press%20release.pdf, accessed 2026-10-20. Translated from German by the case writer. The release names Deilmann, Schwarz and Sawadski as founders, gives the beta test's savings as 12% to 40% "compared with heating behaviour before using tado°", and offers an installation service as an option.
3. tado°, "Häufige Fragen" (FAQ), Internet Archive capture of 18 November 2012, https://web.archive.org/web/20121118050001/http://www.tado.com:80/de/faq.html, accessed 2026-10-20. "Du hast volle Kostenkontrolle durch unser einfaches Pre-paid Modell." Translated from German by the case writer.
4. tado°, FAQ, Internet Archive capture of 8 April 2013, https://web.archive.org/web/20130408101143/http://www.tado.com:80/de/faq.html, accessed 2026-10-20. "Die tado° Basis Hardware gibt es aktuell noch umsonst (eigentlich 149€ einmalig)." "Die tado° Basis-Hardware ist derzeit gratis (sonst 149€ einmalig)." "Die Hardware ist ab dann auch dein Eigentum. Die Basisfunktionen von tado° (manuelle Steuerung) kannst du natürlich weiterhin nutzen." The FAQ also lists the package contents, the box, a gateway and a network cable to the router. Translated from German by the case writer.
5. tado°, landing page LP4, Internet Archive capture of 19 June 2013, https://web.archive.org/web/20130619031522/http://www.tado.com/de/lp4.html, accessed 2026-10-20, read with its page code. The imprint names "Geschäftsführer: Christian Deilmann & Johannes Schwarz & Leopold v. Bismarck". The attached terms say a customer orders with the button "zahlungspflichtig bestellen" "nach Ablauf einer Kompatibilitätsprüfung" (after a compatibility check) and list the payment methods. Translated from German by the case writer.
6. tado°, landing page LP5, Internet Archive capture of 18 June 2013, https://web.archive.org/web/20130618093809/http://www.tado.com/de/lp5.html, accessed 2026-10-20, read with its page code.
7. tado°, landing page LP6, Internet Archive capture of 19 June 2013, https://web.archive.org/web/20130619031850/http://www.tado.com/de/lp6.html, accessed 2026-10-20, read with its page code. Translated from German by the case writer.
8. tado°, landing page LP4, Internet Archive capture of 19 July 2013, https://web.archive.org/web/20130719012322/http://www.tado.com:80/de/lp4.html, accessed 2026-10-20. The visible text matches the capture of 19 June 2013.
9. tado°, "tado° online kaufen", offer page with FAQ, Internet Archive capture of 3 March 2013, https://web.archive.org/web/20130303010127/http://www.tado.com:80/de/angebot.html, accessed 2026-10-19. "Nach Ablauf des ersten Jahres kannst Du die intelligente Regelung von tado° jederzeit monatlich kündigen. Die Kernfunktion um Deine Heizung per App und Webinterface zu steuern bleibt dir auch darüber hinaus erhalten." "Bei einigen Häusern und Wohnungen ist tado° derzeit allerdings noch nicht verfügbar." "Fahre im Bestellprozess fort, um zu prüfen, ab wann tado° für dich in Frage kommt." Translated from German by the case writer.
10. co2online gemeinnützige GmbH, "Bundesweiter Heizspiegel 2012", press release, 22 May 2012, PDF, https://www.heizspiegel.de/fileadmin/pressemitteilungen/120522_PM_BWHS_2012.pdf, Internet Archive copy https://web.archive.org/web/20260209050048/https://www.heizspiegel.de/fileadmin/pressemitteilungen/120522_PM_BWHS_2012.pdf, accessed 2026-10-20.
11. tado°, offer page variant, Internet Archive captures of 16 and 25 March 2013, https://web.archive.org/web/20130316005537/http://www.tado.com:80/de/angebot1.html and https://web.archive.org/web/20130325062150/http://www.tado.com:80/de/angebot1.html, accessed 2026-10-20. "durchschnittlich 27% Energieersparnis (entspricht ca. 285 € pro Jahr)". Translated from German by the case writer.
12. tado°, offer page, Internet Archive capture of 16 June 2013, https://web.archive.org/web/20130616182809/https://www.tado.com/de/angebot.html, accessed 2026-10-20. Translated from German by the case writer.

13. HouseControllers, “tado° – die Nest-Alternative für Deutschland”, 17 February 2013, original post below an update of 6 February 2014, https://www.housecontrollers.de/klima-energie/tado-die-nest-alternative-fuer-deutschland/, accessed 2026-10-19. Translated from German by the case writer.
14. ifun.de, “Erfahrungsbericht: Hausautomatisierung mit RWE SmartHome”, 11 February 2013, https://www.ifun.de/rwe-smarthome-iphone-ipad-mac-test-review-35390, accessed 2026-10-20. Translated from German by the case writer.
15. Simon Tischer, “tado°: Wohltemperiert im Internet der Dinge”, Munich Startup, 22 March 2016, https://munich-startup.de/news/tado-wohltemperiert-im-internet-der-dinge, accessed 2026-10-19. A recollection told in 2016. Translated from German by the case writer.
16. tado°, “A year with tado°”, company blog, 21 December 2012, Internet Archive capture of 11 September 2013, https://web.archive.org/web/20130911000736/http://www.tado.com:80/de-en/blog/a-year-with-tado, accessed 2026-10-19. Around 50 households in the closed beta, “the support of our two investors Target Partners and Shortcut Ventures”, the launch “on November 14th in Germany, Austria and Switzerland”.
17. Christian Deilmann, “ECO12 Berlin: Christian Deilmann Tado Cloud Climate Control”, talk at Ecosummit Berlin 2012, video uploaded 21 June 2012, https://www.youtube.com/watch?v=6oyu0HolO2s, 0:06:22 to 0:07:33, transcribed by the case writer, accessed 2026-10-19.
18. hy!, “Christian Deilmann, tado - hy! January 2013”, video of the pitch and the panel’s feedback, uploaded 24 January 2013, https://www.youtube.com/watch?v=xYCRDtmaQtc, transcribed by the case writer, accessed 2026-10-20. Deilmann 0:03:14 to 0:03:20 and 0:03:38 to 0:03:41. Panelist 1 0:05:02 to 0:05:14 and 0:07:36 to 0:07:52, panelist 2 0:05:44 to 0:05:54, panelist 3 0:06:44 to 0:07:05. Deilmann’s answer 0:07:53 to 0:08:44. Words in the recording that could not be settled are left out of the quotes.
19. tado°, order page “1 Jahr tado° (inkl. Hardwareset)”, Internet Archive capture of 31 January 2013, https://web.archive.org/web/20130131112453/http://www.tado.com:80/de/basispaket.html, accessed 2026-10-20. The page shows the step “Schritt 1/4 System-Auswahl” and the Facebook and Twitter counters. Translated from German by the case writer.
20. tado° and Polarstern, “Zwei junge Münchner Unternehmen heizen dem Energiemarkt ein”, press release, Munich, 21 March 2013, PDF, Internet Archive capture of 24 August 2013, https://web.archive.org/web/20130824072631/http://www.tado.com:80/de/presse.html?&file=tl_files/content/press/releases/21032013_tado_press%20release_Polarstern.pdf, accessed 2026-10-20. Translated from German by the case writer.
21. Robin Wauters, “Tado°, Europe’s answer to NEST, launches its smart heating app worldwide”, The Next Web, 5 March 2013, https://thenextweb.com/news/tado-nest-smart-heating, accessed 2026-10-19.
22. tado°, product page “tado Basispaket A”, Internet Archive capture of 18 November 2012, https://web.archive.org/web/20121118070121/http://www.tado.com:80/de/details/product/tado-Basispaket-A-Raumtemperaturgefuehrt.html, accessed 2026-10-20. “tado° ist ein Service mit einer jährlichen Gebühr.” Translated from German by the case writer.
23. Venture TV, “Tado: Heizkosten senken dank smarter App”, video interview with Christian Deilmann at LeWeb, uploaded 18 February 2013, https://www.youtube.com/watch?v=rytKDwmpr7M, 0:01:56 to 0:02:21, transcribed and translated from German by the case writer, accessed 2026-10-20. In the same interview Deilmann said tado° had been online “since mid-November”. “Wir haben das Produkt im Moment [...] als Service konzipiert, weil wir kontinuierlich für den Kunden arbeiten.” “Wir verlangen eine jährliche Gebühr von 99 Euro.” “Wir schenken dem Kunden die Hardware, damit das Ganze möglich ist.”
24. tado°, FAQ, Internet Archive capture of 8 March 2013, https://web.archive.org/web/20130308110013/http://www.tado.com:80/de/faq.html, accessed 2026-10-20. “Der Vertrag wird zunächst für die Dauer eines Jahres geschlossen. Er verlängert sich danach automatisch auf unbestimmte Zeit und Du kannst monatlich kündigen.”
25. tado°, offer page, Internet Archive capture of 15 March 2013, https://web.archive.org/web/20130315061752/http://www.tado.com:80/de/angebot.html, accessed 2026-10-20. The page code carries a Google Analytics experiment script. Translated from German by the case writer.
26. iphone-ticker.de, “iPhone-gesteuerte Heizung mit tado° – Die deutsche Antwort auf das NEST-Thermostat”, 27 March 2013, with reader comments, Internet Archive capture of 30 June 2013, https://web.archive.org/web/20130630104557/http://www.iphone-ticker.de:80/iphone-gesteuerte-heizung-mit-tado%C2%B0-die-deutsche-antwort-auf-das-nest-thermostat-45385/, accessed 2026-10-20. Comments of 27, 28 and 30 March and 28 June 2013 as posted, the latest in the capture. tado°’s three replies are dated 28 March and 3 April 2013. Translated from German by the case writer.

27. tado°, landing page for Facebook ads, Internet Archive capture of 30 April 2013, https://web.archive.org/web/20130430045345/http://www.tado.com:80/de/lp2.html?utm_source=fbppc_de, accessed 2026-10-20. “Alle Informationen zu Verfügbarkeit und Bestellung erhältst du unter 089 - 416156647 oder [wir] rufen dich kostenfrei zurück”, with name and phone as required fields. The page also showed an image labelled “Frühlingsaktion” (spring promotion) whose content was not captured. Translated from German by the case writer.
28. tado°, job advertisement “Teamleiter Direct Sales / Verkaufsprofi (m/w)”, PDF, Internet Archive capture of 27 June 2013, https://web.archive.org/web/20130627053048/http://www.tado.com:80/tl_files/content/jobs/ausschreibung_direct_sales_teamlead_fest.pdf, accessed 2026-10-20. “Akquise von Neukunden (vor Ort, Telefon, Messen, etc.)”. Translated from German by the case writer.